

POPULATION

- A population can be defined as including all people or items with the characteristic one wishes to understand
- *Because* there is very rarely enough time or money to gather information from everyone or everything in a population, the goal becomes finding a representative sample (or subset) of that population

POPULATION

- *Note that* the population from which the sample is drawn may not be the same as the population about which we actually want information
- Sometimes they may be entirely separate - for instance
 - we might study rats in order to get a better understanding of human health, or
 - we might study records from people born in 2008 in order to make predictions about people born in 2009

SAMPLING

- A sample is “a smaller (but hopefully representative) collection of units from a population used to determine truths about that population” (Field, 2005)
- Why sample?
 - Resources (time, money) and workload
 - Gives results with known accuracy that can be calculated mathematically
- The sampling frame is the list from which the potential respondents are drawn
 - Registrar’s office
 - Class rosters
 - Must assess sampling frame errors

SAMPLING.....

- What is your population of interest?
 - To whom do you want to generalize your results?
 - All doctors
 - School children
 - Indians
 - Women aged 15-45 years
 - Other
- Can you sample the entire population?

SAMPLING.....

- 3 factors that influence sample representative-ness
 - Sampling procedure
 - Sample size
 - Participation (response)

- When might you sample the entire population?
 - When your population is very small
 - When you have extensive resources
 - When you don't expect a very high response

TYPES OF SAMPLES

➤ Probability (Random) Samples

❖ Simple random sample

- Systematic random sample
- Stratified random sample
- Multistage sample
- Multiphase sample
- Cluster sample

➤ Non-Probability Samples

- Convenience sample
- Purposive sample
- Quota

PROBABILITY SAMPLING

- A **probability sampling** scheme is one in which every unit in the population has a chance (greater than zero) of being selected in the sample, and this probability can be accurately determined.
- . When every element in the population *does* have the same probability of selection, this is known as an 'equal probability of selection' (EPS) design. Such designs are also referred to as 'self-weighting' because all sampled units are given the same weight.

SIMPLE RANDOM SAMPLING

- Applicable when population is small, homogeneous and readily available
- All subsets of the frame are given an equal probability. Each element of the frame thus has an equal probability of selection.
- It provides for greatest number of possible samples. This is done by assigning a number to each unit in the sampling frame.
- A table of random number or lottery system is used to determine which units are to be selected.

SIMPLE RANDOM SAMPLING.....

- Estimates are easy to calculate
- Simple random sampling is always an EPS design, but not all EPS designs are simple random sampling.

Disadvantages

- If sampling frame large, this method impracticable
- Minority subgroups of interest in population may not be present in sample in sufficient numbers for study.

REPLACEMENT OF SELECTED UNITS

- Sampling schemes may be *without replacement* ('WOR' - no element can be selected more than once in the same sample) or *with replacement* ('WR' - an element may appear multiple times in the one sample)
- For example, if we catch fish, measure them, and immediately return them to the water before continuing with the sample, this is a WR design, because we might end up catching and measuring the same fish more than once. However, if we do not return the fish to the water (e.g. if we eat the fish), this becomes a WOR design.

SYSTEMATIC SAMPLING

- Systematic sampling relies on arranging the target population according to some ordering scheme and then selecting elements at regular intervals through that ordered list
- Systematic sampling involves a random start and then proceeds with the selection of every k th element from then onwards. In this case, $k = (\text{population size} / \text{sample size})$
- It is important that the starting point is not automatically the first in the list, but is instead randomly chosen from within the first to the k th element in the list
- A simple example would be to select every 10th name from the telephone directory (an 'every 10th' sample, also referred to as 'sampling with a skip of 10').

SYSTEMATIC SAMPLING...

As described above, systematic sampling is an EPS method, because all elements have the same probability of selection (in the example given, one in ten). It is *not* 'simple random sampling' because different subsets of the same size have different selection probabilities - e.g. the set $\{4, 14, 24, \dots, 994\}$ has a one-in-ten probability of selection, but the set $\{4, 13, 24, 34, \dots\}$ has zero probability of selection.



SYSTEMATIC SAMPLING.....

Advantages

- Sample easy to select
- Suitable sampling frame can be identified easily
- Sample evenly spread over entire reference population

Disadvantages

- Sample may be biased if hidden periodicity in population coincides with that of selection.
- Difficult to assess precision of estimate from one survey.

STRATIFIED SAMPLING

- Where population embraces a number of distinct categories, the frame can be organized into separate "strata"
- Each stratum is then sampled as an independent sub-population, out of which individual elements can be randomly selected
- Every unit in a stratum has same chance of being selected
- Using same sampling fraction for all strata ensures proportionate representation in the sample
- Adequate representation of minority subgroups of interest can be ensured by stratification and varying sampling fraction between strata as required

STRATIFIED SAMPLING.....

- Finally, since each stratum is treated as an independent population, different sampling approaches can be applied to different strata.

Drawbacks to using stratified sampling

- First, sampling frame of entire population has to be prepared separately for each stratum
- Second, when examining multiple criteria, stratifying variables may be related to some, but not to others, further complicating the design, and potentially reducing the utility of the strata
- Finally, in some cases (such as designs with a large number of strata, or those with a specified minimum sample size per group), stratified sampling can potentially require a larger sample than would other methods